Tertiary Infotech Academy Pte Ltd

Lab 04 — Model a Portfolio with Python Collections

Python Programming for Finance · TGS-2026065049 · LO2

Scenario

An investment team needs a clean in-memory representation of holdings, identifiers, sector groups and prices.

Goal

A portfolio report using list, tuple, dictionary and set operations.

Tools

Python, csv

Step-by-step

1. Inspect portfolio.csv and identify suitable Python types.

2. Load rows into a list of dictionaries.

```
python starter.py
```

3. Represent the fixed instrument key as a tuple.

4. Build ticker-to-price and ticker-to-sector dictionaries.

5. Use sets to calculate overlap between two portfolios.

6. Print market value and weights; confirm weights total 100%.

```
python solution.py
```

Acceptance criteria

Every holding has a market value, portfolio weights sum to 100%, and the overlap set is reported without duplicates.

Evidence

Capture the executed output, validation result, generated file or chart, and a one-sentence finance interpretation.